CINCINNATI TRANSIT AUTHORITY 2026

FOH CHANNEL PROCESSING — DiGiCo Quantum 225

PLUGIN: Waves SSL EV2 Channel Strip (SSL 4000E Emulation) — per channel

VENUE: Memorial Hall ('Memo') — Cincinnati, OH — RT60: ~2.2s — Capacity: 556 seats

EQ BANDS: High → Low Order | Thresholds: Calibrated for -25 to -20 dBFS Input Metering

SSL EV2 Channel Strip — Key Concepts for This Session:
EQ Types: Brown Knob (O2) = wider, grittier, musical — excellent for drums, bass, horns. Black Knob (242) = cleaner, narrower Q — excellent for vocals, acoustic, keys.
Filter Section: Sweepable HPF and LPF — placed before or after dynamics depending on SPLIT routing setting. **Default routing:** Dyn pre-EQ (compressor sees raw signal; EQ shapes afterward).
Analog Saturation: ON adds THD to input and output stages (mic pre emulation). OFF for clean sources. Driving the input (+2 to +5 dB) increases harmonic richness.
Gate/Expander: Uses same gain-change circuit as comp. 'Range' sets maximum attenuation. F.ATK button = fast compressor attack for transient control (drums, bass).
Auto Makeup: EV2 automatically calculates makeup gain from ratio + threshold; add Manual Makeup on top.

INPUT LIST SUMMARY — Cincinnati Transit Authority 2026

Ch	Instrument	Mic / DI	EV2 EQ Type
1	Kick	Earthworks DM6 SeisMic	Brown Knob
2	Snare	Lauten Audio Snare Mic (FET Condenser)	Brown Knob
3	Hi-Hat	AKG C451	Black Knob
4	Rack Tom	Earthworks DM17	Brown Knob
5	Floor Tom 1	Earthworks DM17	Brown Knob
6	Floor Tom 2	Earthworks DM17	Brown Knob
7	OH Left	Earthworks SR20	Black Knob
8	OH Right	Earthworks SR20	Black Knob
9	Knee Mic (Shells)	12 Gauge Gold12 (Shotgun)	Brown Knob
10	Tracks (Playback)	DI / XLR	Black Knob
11	Bass DI	DI (Active)	Brown Knob
12	Bass AMP	Earthworks DM6 SeisMic	Brown Knob
13	Guitar Left	DI / Amp (XLR)	Brown Knob
14	Guitar Right	DI / Amp (XLR)	Brown Knob
15	Acoustic Guitar	DI (Active)	Black Knob
16	Trombone	Shure Beta 98H/C	Black Knob
17	Trumpet	Shure Beta 98H/C	Black Knob
18	Saxophone	Shure Beta 98H/C	Black Knob
19	Flute	DPA 4099 (d:vote CORE)	Black Knob
20	Keys 1	DI (Active)	Black Knob
21	Keys 2	DI (Active)	Black Knob

22	Keys Vocal	Shure Beta 58	Black Knob
23	Bass Vocal	Shure Beta 58	Black Knob
24	Drum Vocal	Shure Beta 58	Black Knob
25	Guitar Vocal	Shure Beta 58	Black Knob
26	Star Vocal (Lead)	Shure Beta 58	Black Knob

Ch 1 — Kick | Earthworks DM6 SeisMic | EQ: Brown Knob

Section	Parameter	Details
INPUT	Analog Sat	ON — Line position; drive lightly (+2 to +4 dB input) for console-style THD on kick transients
FILTER	LPF	8 kHz — Rolls off harsh upper content the room smears; DM6 is a condenser so upper-freq detail in a 2.2s RT60 creates wash
FILTER	HPF	60 Hz — Aggressive sub cut; Memo's low-freq buildup at 556-seat room modes. DM6 captures full-range 20Hz; cut below usable bass thump
EQ — HF	HF Shelf	8 kHz / -2 dB — Gentle shelf trim to complement LPF, tames any pre-ringing from condenser attack in the room
EQ — HMF	HMF Bell	3.5 kHz / +3 dB / Q 0.7 — Adds beater attack and click through a dense horn section mix; Brown Knob Q behavior makes this musical
EQ — LMF	LMF Bell	300 Hz / -4 dB / Q 0.8 — PRIMARY mud cut. Memo's 2.2s RT60 blooms this range aggressively; keep kick punchy not boomy
EQ — LF	LF Bell	80 Hz / +3 dB / Q 0.5 — Adds thump and sub-body. Brown Knob wide Q works beautifully here for kick fundamental, not peaked
COMP	Threshold	-22 dBFS — Calibrated for your -25 to -20 dBFS input metering; compressor engages on every hit
COMP	Ratio	4:1
COMP	Attack	FAST (F.ATK engaged) — Tightens transient, gives SSL 'smack' characteristic to beater click
COMP	Release	~250 ms (program-sensitive auto release active) — Let kick breathe between hits at tempo
COMP	Makeup Gain	+3 dB — Compensates for 4:1 at -22 dBFS threshold; auto makeup also active
COMP	Routing	Dyn pre-EQ (default) — Compressor sees raw DM6 signal; then EQ shapes tonal character
GATE	Threshold	-38 dBFS — Below snare/tom bleed but above silence; DM6 supercardioid helps with isolation
GATE	Range	-30 dB — Partial gate, not full close; preserves natural kick shell resonance
GATE	Attack	FAST — Instant open on beater impact
GATE	Release	200 ms — Allows ring to breathe before close

Engineer Notes: *The DM6 SeisMic is a condenser that delivers a near-flat, pre-EQ'd kick sound requiring minimal surgery — but Memo's 2.2s RT60 demands we be more aggressive than in a dry room. The 300 Hz cut (LMF) is the most critical move, pulling mud that will bloom into the room's long decay. Brown Knob EQ gives the 80 Hz body and 3.5 kHz click a musical, wide-Q SSL character. SSL 'smack' from fast attack compressor tightens the beater without over-squashing. Gate range of -30 dB (not full close) preserves natural shell resonance.*

Ch 2 — Snare | Lauten Audio Snare Mic (FET Condenser) | EQ: Brown Knob

Section	Parameter	Details
INPUT	Analog Sat	ON — Line position; slight THD warmth helps FET condenser avoid sterile thin sound in the room
MIC NOTE	Onboard HPF	Set mic body switch to 80 Hz — Removes kick drum bleed at source before reaching EV2. Onboard 140 Hz is too aggressive for CTA snare tone
MIC NOTE	Onboard LPF	Set to FLAT — Let EV2 handle high-end shaping; Lauten 12kHz position creates dip not needed here
FILTER	LPF	12 kHz — Rolls off hi-hat shimmer that bleeds into snare mic in this live environment
FILTER	HPF	100 Hz — Plugin HPF supplements mic body switch; combined 80 Hz mic + 100 Hz plugin = clean low-end rejection
EQ — HF	HF Shelf	10 kHz / +2 dB — Air and crack; Lauten naturally has a Lauten-voiced dip at 3 kHz, we restore presence here
EQ — HMF	HMF Bell	5 kHz / +3 dB / Q 1.0 — Stick impact and crack through the mix; Lauten's freq contour has 3kHz dip so boost at 5k gives snap
EQ — LMF	LMF Bell	250 Hz / -5 dB / Q 0.8 — Aggressive Memo mud cut; snare ring in reverberant rooms creates 200-300 Hz buildup that clutters the horn section
EQ — LF	LF Bell	100 Hz / +1.5 dB / Q 0.5 — Subtle body boost; snare needs some chest weight without competing with kick at 80 Hz
COMP	Threshold	-20 dBFS — Engages on hard hits, light compression on ghost notes
COMP	Ratio	3:1
COMP	Attack	SLOW (F.ATK disengaged) — Let transient through for crack, then compress the body
COMP	Release	~150 ms auto — Fast enough to recover between hits at up-tempo Chicago-style rhythms
COMP	Makeup Gain	+2 dB
COMP	Routing	Dyn pre-EQ (default)
GATE	Threshold	-32 dBFS — Tighter than kick gate; snare mic picks up kit
GATE	Range	-20 dB — Partial close; preserves snare ring between hits
GATE	Attack	FAST
GATE	Release	150 ms

Engineer Notes: *The Lauten Snare Mic's FET condenser character has a natural dip around 3 kHz and extended high-end — quite different from an SM57. Our EV2 approach compensates with the 5 kHz HMF boost for crack and a 10 kHz air shelf. The 250 Hz LMF cut at -5 dB is the key Memo-specific move: snare ring blooms massively in a 2.2s space and creates mud in the 200-300 Hz range that fights the trombone. Slow compressor attack lets the crack speak before taming the body, giving Chicago-style authority to the backbeat.*

Ch 3 — Hi-Hat | AKG C451 | EQ: Black Knob

Section	Parameter	Details
INPUT	Analog Sat	OFF — Hi-hat is a detail mic; keep it clean with no THD saturation
FILTER	LPF	OFF — Let hi-hat breathe naturally up top
FILTER	HPF	250 Hz — Aggressive! 451 on hat in live setting picks up enormous kick and snare low-end; cut it all
EQ — HF	HF Shelf	10 kHz / -2 dB — C451 has a famous presence peak; tame it slightly for Memo so it doesn't spray shimmer into the room reverb
EQ — HMF	HMF Bell	5 kHz / -2 dB / Q 1.5 — 451 harshness zone; small cut cleans up the metallic edge that gets ugly in reverberant rooms
EQ — LMF	LMF Bell	FLAT — Nothing useful below 500 Hz; HPF handles it
EQ — LF	LF	FLAT
COMP	Threshold	-18 dBFS — Light compression, just controlling peaks
COMP	Ratio	2:1
COMP	Attack	SLOW — Don't compress the attack transient of hat
COMP	Release	~100 ms auto
COMP	Makeup Gain	+1 dB
GATE	Threshold	-28 dBFS — Keep gate tight; 451 hears everything on kit
GATE	Range	-40 dB — Hard close; hat is punctuation only
GATE	Attack	FAST
GATE	Release	80 ms

Engineer Notes: *The AKG C451 on hi-hat is a supporting detail mic — this is NOT a featured element. The 250 Hz HPF is intentionally aggressive because in Memo's room the bleed from kick and snare into the hat mic creates low-end mush. The 10kHz shelf cut tames the C451's famous presence peak that would otherwise spray into the room's 2.2s decay and smear the mix. Keep this channel lower in the mix than you think — the overheads cover the kit's open sound.*

Ch 4 — Rack Tom | Earthworks DM17 | EQ: Brown Knob

Section	Parameter	Details
INPUT	Analog Sat	ON — Light THD adds warmth to the DM17's clean condenser character on toms
FILTER	LPF	10 kHz — DM17 is very extended; roll off upper content that gets smeared in the room
FILTER	HPF	80 Hz — Remove sub rumble; rack tom fundamental starts around 100-180 Hz
EQ — HF	HF Shelf	8 kHz / -1.5 dB — Tame attack shimmer in reverberant room
EQ — HMF	HMF Bell	4 kHz / +2.5 dB / Q 0.8 — Attack click and stick definition through the mix
EQ — LMF	LMF Bell	300 Hz / -4 dB / Q 0.7 — Mud control; Memo RT60 blooms this range on toms
EQ — LF	LF Bell	120 Hz / +3 dB / Q 0.6 — Tom body and thump; Brown Knob width works well here
COMP	Threshold	-22 dBFS
COMP	Ratio	4:1
COMP	Attack	FAST — Tom attack is the key sonic element
COMP	Release	200 ms auto
COMP	Makeup Gain	+3 dB
GATE	Threshold	-35 dBFS
GATE	Range	-30 dB — Partial close, preserves ring
GATE	Attack	FAST
GATE	Release	250 ms — Longer release for tom decay to sustain naturally

Engineer Notes: *The DM17 is a small cardioid condenser purpose-built for toms — very fast transient response and extended response. In Memo, we need to pull the 300 Hz range aggressively to keep the rack tom from becoming a boomy smear in the room's decay. The Brown Knob's wider Q at the 120 Hz LF boost gives the tom body that feels natural rather than peaked. Extended gate release of 250 ms allows the characteristic tom ring to decay naturally before the gate closes.*

Ch 5 — Floor Tom 1 | Earthworks DM17 | EQ: Brown Knob

Section	Parameter	Details
INPUT	Analog Sat	ON — Light saturation on floor tom adds low-end warmth
FILTER	LPF	10 kHz
FILTER	HPF	60 Hz — Floor tom sits lower than rack; keep more sub
EQ — HF	HF Shelf	8 kHz / -1.5 dB
EQ — HMF	HMF Bell	3.5 kHz / +2.5 dB / Q 0.8 — Attack click
EQ — LMF	LMF Bell	250 Hz / -4 dB / Q 0.8 — Mud; floor tom builds at 200-300 Hz in reverberant rooms
EQ — LF	LF Bell	90 Hz / +3 dB / Q 0.5 — Deep body, sits lower than rack tom
COMP	Threshold	-22 dBFS
COMP	Ratio	4:1
COMP	Attack	FAST
COMP	Release	250 ms auto
COMP	Makeup Gain	+3 dB
GATE	Threshold	-33 dBFS
GATE	Range	-28 dB
GATE	Attack	FAST
GATE	Release	300 ms — Floor tom has the most natural ring of all toms; let it breathe

Engineer Notes: *Floor Tom 1 uses the same DM17 as the rack tom but sits tuned lower. The HPF is slightly looser at 60 Hz to preserve the deeper fundamental, and the LF boost is at 90 Hz rather than 120 Hz. The 250 Hz mud cut is essential — floor toms can dominate the low-mid range in a reverberant room like Memo. Gate release of 300 ms allows the floor tom's characteristic extended ring to sustain naturally.*

Ch 6 — Floor Tom 2 | Earthworks DM17 | EQ: Brown Knob

Section	Parameter	Details
INPUT	Analog Sat	ON
FILTER	LPF	10 kHz
FILTER	HPF	55 Hz — Deepest floor tom; preserve low fundamental
EQ — HF	HF Shelf	8 kHz / -1.5 dB
EQ — HMF	HMF Bell	3.5 kHz / +2 dB / Q 0.8
EQ — LMF	LMF Bell	240 Hz / -4 dB / Q 0.8 — Mud control
EQ — LF	LF Bell	80 Hz / +3 dB / Q 0.5 — Deepest body boost
COMP	Threshold	-22 dBFS
COMP	Ratio	4:1
COMP	Attack	FAST
COMP	Release	280 ms auto
COMP	Makeup Gain	+3 dB
GATE	Threshold	-33 dBFS
GATE	Range	-28 dB
GATE	Attack	FAST
GATE	Release	320 ms

Engineer Notes: *Floor Tom 2 is the deepest drum on the kit — tuned down from Floor Tom 1. HPF at 55 Hz and LF boost at 80 Hz preserve the deepest fundamental. Same mud-control strategy as the other toms but centered slightly lower at 240 Hz. Match this channel closely to Floor Tom 1 in terms of level and decay character so the toms speak as a unified section through fills.*

Ch 7 — OH Left | Earthworks SR20 | EQ: Black Knob

Section	Parameter	Details
INPUT	Analog Sat	OFF — SR20 is an ultra-clean small-diaphragm condenser; preserve its transparency
FILTER	LPF	OFF — Need full overhead air
FILTER	HPF	120 Hz — Remove low-end kit rumble from overheads; close mics handle kick/snare body
EQ — HF	HF Shelf	12 kHz / -2 dB — SR20 is extended; tame slightly to prevent the room reverb from building harsh top-end spray
EQ — HMF	HMF Bell	3 kHz / -2 dB / Q 1.5 — Small cut reduces SR20's slight presence lift that can sound harsh in the room
EQ — LMF	LMF Bell	200 Hz / -3 dB / Q 0.8 — Low-mid mud that overheads pick up from the full kit in the room
EQ — LF	LF	FLAT — HPF handles the low end
COMP	Threshold	-16 dBFS — Gentle compression; overheads should breathe
COMP	Ratio	2:1
COMP	Attack	SLOW — Don't compress the cymbal transients; they define the kit's spatial presence
COMP	Release	~400 ms auto — Long release keeps overheads open-sounding
COMP	Makeup Gain	+1 dB
GATE	State	OFF — Never gate overheads; they provide the room glue for the kit

Engineer Notes: *The Earthworks SR20 overheads are ultra-linear small-diaphragm condensers with extended response — they'll tell you exactly what the room sounds like, which in Memo's case includes that 2.2s reverb. Black Knob EQ is the right choice here: cleaner, narrower corrections that preserve the SR20's transparency. The 120 Hz HPF prevents low-end buildup while keeping the full cymbal spectrum intact. Critical note: keep overhead level conservative in Memo — the room adds natural ambience. DO NOT gate overheads.*

Ch 8 — OH Right | Earthworks SR20 | EQ: Black Knob

Section	Parameter	Details
INPUT	Analog Sat	OFF
FILTER	LPF	OFF
FILTER	HPF	120 Hz
EQ — HF	HF Shelf	12 kHz / -2 dB
EQ — HMF	HMF Bell	3 kHz / -2 dB / Q 1.5
EQ — LMF	LMF Bell	200 Hz / -3 dB / Q 0.8
EQ — LF	LF	FLAT
COMP	Threshold	-16 dBFS
COMP	Ratio	2:1
COMP	Attack	SLOW
COMP	Release	~400 ms auto
COMP	Makeup Gain	+1 dB
GATE	State	OFF
PHASE	Check	Verify L/R phase alignment with Smaart v9 — SR20 pair should be time-aligned for mono compatibility

Engineer Notes: *OH Right is matched exactly to OH Left (see CH 7). The additional note here is phase alignment: with Earthworks SR20s in XY or ORTF configuration over the kit, verify alignment in Smaart v9 before the show. Phase issues between the pair and the close mics (especially kick and snare) will cause low-end smearing that's amplified by the room's long decay.*

Ch 9 — Knee Mic (Shells) | 12 Gauge Gold12 (Shotgun) | EQ: Brown Knob

Section	Parameter	Details
INPUT	Analog Sat	ON — Light THD; shotgun ambient mic benefits from analog character
NOTE	Usage	Ambient shell/room mic positioned at knee level — captures kit size and low-end room interaction. Use sparingly in Memo; room already adds reverb naturally
FILTER	LPF	6 kHz — This is an ambient/blend mic; cut high content hard to avoid adding more room shimmer on top of Memo's own RT60
FILTER	HPF	120 Hz — Cut sub and low-end rumble from the floor
EQ — HF	HF Shelf	5 kHz / -3 dB — Heavy high cut; this mic's job is body, not detail
EQ — HMF	HMF Bell	FLAT
EQ — LMF	LMF Bell	300 Hz / -3 dB / Q 0.8 — Mud control even on ambient mic
EQ — LF	LF Bell	100 Hz / +2 dB / Q 0.5 — Shell resonance and room body
COMP	Threshold	-20 dBFS
COMP	Ratio	3:1
COMP	Attack	SLOW — Ambient mic; let the room breathe through it
COMP	Release	500 ms auto — Long release; this is a blend/glue element
COMP	Makeup Gain	+2 dB
GATE	State	OFF — Ambient mic should be open
MEMO NOTE	Level	Keep this fader -6 to -10 dB below overheads; do NOT use in main PA mix without significant experimentation in this room

***Engineer Notes:** The knee/shells mic in Memorial Hall needs to be approached with extreme caution. Memo's 2.2s RT60 already provides abundant natural room sound — adding a dedicated room mic can quickly make the kit sound washy. Treat this channel as a blending tool at low levels for body and shell resonance only. Heavy high-end cutting (LPF at 6 kHz, HF shelf at 5 kHz) prevents this mic from adding high-frequency room wash. If in doubt, mute it and trust the room.*

Ch 10 — Tracks (Playback) | DI / XLR | EQ: Black Knob

Section	Parameter	Details
INPUT	Analog Sat	OFF — Playback tracks should be clean and transparent
FILTER	LPF	OFF
FILTER	HPF	40 Hz — Remove subsonic content only
EQ — HF	HF Shelf	12 kHz / +1 dB — Small air boost helps digital playback breathe in the room
EQ — HMF	HMF Bell	FLAT
EQ — LMF	LMF Bell	200 Hz / -2 dB / Q 0.8 — Small Memo mud cut on playback low-mid
EQ — LF	LF Shelf	60 Hz / -2 dB — Gently reduce sub that the room will amplify
COMP	Threshold	-12 dBFS — Light limiting/compression; tracks are already mastered
COMP	Ratio	2:1 — Gentle bus-style compression
COMP	Attack	SLOW
COMP	Release	500 ms auto
COMP	Makeup Gain	0 dB — Tracks set own level
GATE	State	OFF — Never gate playback
NOTE	Stereo	Run as stereo linked pair if console permits; tracks are stereo content

Engineer Notes: *Playback tracks from a digital playback rig are already mixed and mastered — minimal processing is the goal here. The SSL EV2 provides gentle glue and subtle room-adaptation: a small 200 Hz cut prevents the playback low-mids from blooming in Memo's decay. Keep the compressor very gentle (2:1) as a safety limiter only. The 40 Hz HPF removes any subsonic content that would excite Memo's room modes.*

Ch 11 — Bass DI | DI (Active) | EQ: Brown Knob

Section	Parameter	Details
INPUT	Analog Sat	ON — Drive at +3 to +5 dB; the Brown Knob + analog saturation on bass DI is THE classic SSL low-end sound
FILTER	LPF	5 kHz — DI bass has full spectrum; roll off upper content that creates harshness in the room
FILTER	HPF	40 Hz — Remove subsonic; keep usable bass
EQ — HF	HF Shelf	4 kHz / +2 dB — String definition and pick/slap attack
EQ — HMF	HMF Bell	800 Hz / -2.5 dB / Q 0.8 — Bass honk zone; cut removes midrange mud
EQ — LMF	LMF Bell	300 Hz / -4 dB / Q 0.8 — Memo mud control; bass low-mids bloom aggressively in this room
EQ — LF	LF Bell	80 Hz / +3 dB / Q 0.5 — Bass fundamental; Brown Knob wide Q is perfect for adding musical sub without boxiness
COMP	Threshold	-22 dBFS
COMP	Ratio	4:1
COMP	Attack	FAST — SSL fast attack is classic for DI bass; controls pick attack
COMP	Release	~200 ms auto
COMP	Makeup Gain	+4 dB
COMP	Routing	Dyn pre-EQ — Compress the raw DI signal, then EQ shapes tonality
GATE	State	OFF — Bass should sustain naturally between notes

Engineer Notes: *Bass DI through the SSL EV2 with Brown Knob EQ and analog saturation engaged is a legendary combination — it's the foundation of the 'SSL bass sound' that defined 80s and 90s records. The CTA tribute band playing Chicago material needs a round, present, well-defined bass that supports the horn section without fighting it in Memo's low-mid buildup. The 300 Hz cut is critical: this is where Chicago's horn section and the bass compete most directly. Keep the DI channel slightly back in the mix and blend with the Amp mic (CH 12) for natural tone.*

Ch 12 — Bass AMP | Earthworks DM6 SeisMic | EQ: Brown Knob

Section	Parameter	Details
INPUT	Analog Sat	ON — Heavy saturation here; drive the DM6 into the input for amp-like harmonic distortion
NOTE	DM6 on Amp	DM6 on bass cab is its 'secret weapon' application per Earthworks engineers; condenser capsule captures amp character with fast transient and full extension
FILTER	LPF	4 kHz — Bass amp is about body, not detail; hard roll-off above 4k
FILTER	HPF	50 Hz — Remove subsonic from amp
EQ — HF	HF Shelf	3 kHz / -2 dB — Roll off amp brightness further
EQ — HMF	HMF Bell	1.2 kHz / +2 dB / Q 0.8 — Amp speaker resonance and cabinet character
EQ — LMF	LMF Bell	350 Hz / -3.5 dB / Q 0.7 — Memo mud cut; amp sounds boxy in this room
EQ — LF	LF Bell	100 Hz / +3 dB / Q 0.5 — Amp body; sits slightly higher than DI sub
COMP	Threshold	-20 dBFS
COMP	Ratio	3:1
COMP	Attack	SLOW — Amp already has natural compression; let it breathe first
COMP	Release	~300 ms auto
COMP	Makeup Gain	+2 dB
GATE	State	OFF
BLEND	DI vs Amp	Blend DI at 70% / AMP at 30% — DI provides clarity and punch; amp provides warmth and character. Adjust by ear during soundcheck

Engineer Notes: *The DM6 on bass amp is a surprisingly excellent combination — Earthworks designed this mic specifically knowing engineers use it on bass cabs. Its condenser capsule captures the cabinet's character with extended frequency response that a dynamic can't match. The EV2 Brown Knob + heavy analog saturation on the amp channel gives the bass section vintage SSL warmth. Blend mostly DI with amp underneath for the best of both worlds — clarity and character.*

Ch 13 — Guitar Left | DI / Amp (XLR) | EQ: Brown Knob

Section	Parameter	Details
INPUT	Analog Sat	ON — Drive input for amp-like saturation; Brown Knob + saturation = classic SSL guitar chain
NOTE	Signal	XLR = DI or load box from guitar amp; treat as direct amp signal
FILTER	LPF	6 kHz — Chicago rhythm guitar is rhythm section; cut harsh upper content that fights horns
FILTER	HPF	100 Hz — Remove low-end that fights kick and bass
EQ — HF	HF Shelf	5 kHz / -1.5 dB — Tame presence spike that bleeds into Memo's reverb
EQ — HMF	HMF Bell	2.5 kHz / +2 dB / Q 1.0 — Guitar cut-through presence; Chicago rhythm guitar needs to be heard under the horns
EQ — LMF	LMF Bell	400 Hz / -3.5 dB / Q 0.8 — Guitar mud/boxiness in reverberant room; critical cut
EQ — LF	LF Bell	150 Hz / +1.5 dB / Q 0.6 — Warm body without competing with bass
COMP	Threshold	-20 dBFS
COMP	Ratio	4:1 — Guitar needs even dynamics under the horn section
COMP	Attack	FAST — Control strumming transients
COMP	Release	~200 ms auto
COMP	Makeup Gain	+3 dB
GATE	State	OFF — Guitar should sustain; use HPF to handle bleed

***Engineer Notes:** Chicago-style rhythm guitar needs to sit in the mix as a rhythmic foundation under a six-piece horn section — not a featured lead instrument. The SSL EV2 Brown Knob + saturation gives the DI guitar signal analog amp character. The 400 Hz cut is essential in Memo: guitar low-mids in a reverberant room create a boxy sound that competes directly with the horn section's 250-500 Hz fundamental range. Keep Guitar Left slightly behind Guitar Right in the stereo field.*

Ch 14 — Guitar Right | DI / Amp (XLR) | EQ: Brown Knob

Section	Parameter	Details
INPUT	Analog Sat	ON
FILTER	LPF	6 kHz
FILTER	HPF	100 Hz
EQ — HF	HF Shelf	5 kHz / -1.5 dB
EQ — HMF	HMF Bell	3 kHz / +2 dB / Q 1.0 — Slightly higher than Guitar Left for stereo differentiation
EQ — LMF	LMF Bell	400 Hz / -3.5 dB / Q 0.8
EQ — LF	LF Bell	150 Hz / +1.5 dB / Q 0.6
COMP	Threshold	-20 dBFS
COMP	Ratio	4:1
COMP	Attack	FAST
COMP	Release	~200 ms auto
COMP	Makeup Gain	+3 dB
GATE	State	OFF
PAN NOTE	Stereo Width	Pan Guitar Left to ~10 o'clock; Guitar Right to ~2 o'clock — creates guitar width without competing in the center where horns live

Engineer Notes: *Guitar Right is matched to Guitar Left with a slightly higher HMF center frequency (3 kHz vs 2.5 kHz) for subtle stereo differentiation in the upper-mid range. The panning note is critical for a horn band: keep both guitars away from center so the trumpet, trombone, and sax can occupy the sonic center stage. Both guitars should feel like a wide rhythmic foundation, not a featured element.*

Ch 15 — Acoustic Guitar | DI (Active) | EQ: Black Knob

Section	Parameter	Details
INPUT	Analog Sat	ON — Light drive (+1 to +2 dB); acoustic DI benefits from subtle THD to add warmth
FILTER	LPF	OFF — Let acoustic air come through
FILTER	HPF	120 Hz — Remove low-end boominess from acoustic pickup/DI
EQ — HF	HF Shelf	10 kHz / +1.5 dB — Air and sparkle; acoustic guitar needs breath above the horn section
EQ — HMF	HMF Bell	2.5 kHz / +2 dB / Q 1.5 — Acoustic string definition and presence; narrower Black Knob Q is precise here
EQ — LMF	LMF Bell	350 Hz / -4 dB / Q 1.0 — Acoustic DI boxiness/proximity effect buildup in reverberant room
EQ — LF	LF Bell	180 Hz / +1.5 dB / Q 0.8 — Warmth without boom
COMP	Threshold	-18 dBFS
COMP	Ratio	3:1
COMP	Attack	SLOW — Let acoustic transient speak; strum attacks define the rhythm
COMP	Release	~300 ms auto
COMP	Makeup Gain	+2 dB
GATE	State	OFF

Engineer Notes: *Acoustic guitar DI through the EV2 with Black Knob EQ and light saturation gives the acoustic signal warmth without compromising its natural character. Black Knob is the right choice over Brown here — the acoustic guitar needs precise, clean corrections rather than the Brown Knob's wider, grittier reshaping. The 350 Hz cut is the most important move: acoustic DI systems are notorious for this frequency range sounding boxy, and Memo amplifies it. The 10 kHz air boost helps the acoustic breathe above the horn section.*

Ch 16 — Trombone | Shure Beta 98H/C | EQ: Black Knob

Section	Parameter	Details
INPUT	Analog Sat	OFF — Beta 98H has tailored frequency response for brass; keep it clean
FILTER	LPF	OFF — Trombone needs full harmonic content through ~8 kHz
FILTER	HPF	80 Hz — Remove sub rumble from trombone; fundamental starts ~75-80 Hz for low Bb
EQ — HF	HF Shelf	10 kHz / -1.5 dB — Trombone can be bright and brassy in the room; small trim prevents harshness in Memo's reverb
EQ — HMF	HMF Bell	2.5 kHz / +2.5 dB / Q 1.2 — Trombone presence and cut-through; Black Knob precision works beautifully here for mid presence
EQ — LMF	LMF Bell	300 Hz / -3.5 dB / Q 0.8 — Trombone low-mid bloom in reverberant room; pull this to keep the horn clear
EQ — LF	LF Bell	120 Hz / +2 dB / Q 0.6 — Trombone fundamental warmth and weight
COMP	Threshold	-20 dBFS
COMP	Ratio	3:1 — Control trombone dynamics which can vary enormously
COMP	Attack	SLOW — Let the brass attack through; attack transient is musical
COMP	Release	~250 ms auto
COMP	Makeup Gain	+2 dB
COMP	Routing	Dyn pre-EQ — Compress the raw horn dynamics first
GATE	State	OFF — Live brass mics; use HPF to manage bleed, not gates

Engineer Notes: *The Beta 98H/C on trombone is a classic combination — Shure specifically voiced this mic for brass instruments with a carefully tailored frequency response. Black Knob EQ gives precise, clean corrections that respect the Beta 98H's natural tonal shaping. The 300 Hz cut is critical: in Memo's 2.2s RT60, trombone low-mids bloom in the room and create mud that clashes with bass guitar and keyboards. Keep trombone slightly forward in the horn section mix — it provides the foundation for the CTA horn sound.*

Ch 17 — Trumpet | Shure Beta 98H/C | EQ: Black Knob

Section	Parameter	Details
INPUT	Analog Sat	OFF
FILTER	LPF	OFF — Trumpet harmonics extend to 12 kHz+; preserve them
FILTER	HPF	150 Hz — Trumpet fundamental starts at ~165 Hz (low E); remove everything below
EQ — HF	HF Shelf	10 kHz / -2 dB — Trumpet can be extremely bright; significant taming prevents the room reverb from making it piercing
EQ — HMF	HMF Bell	3 kHz / +3 dB / Q 1.5 — Trumpet presence and cut — this is where the 'Chicago trumpet sound' lives; Black Knob Q precision is ideal
EQ — LMF	LMF Bell	350 Hz / -3 dB / Q 0.8 — Low-mid mud cut; trumpet in a reverberant room picks up body that muddies the high end
EQ — LF	LF Bell	FLAT — Trumpet fundamental is above 150 Hz; HPF handles the rest
COMP	Threshold	-18 dBFS — Trumpet players can be the wildest dynamic source in a live band
COMP	Ratio	4:1 — More aggressive than trombone; trumpet dynamics are more extreme
COMP	Attack	SLOW — Brass attack is musical; let it through
COMP	Release	~200 ms auto
COMP	Makeup Gain	+3 dB
GATE	State	OFF

Engineer Notes: *Trumpet through the Beta 98H/C needs the most dynamic control of the horn section — trumpet players have enormous dynamic range and in a 556-seat reverberant room, an unrestricted trumpet can dominate the entire mix. The 4:1 ratio is necessary. The 10 kHz shelf trim at -2 dB is critical: Memo's long RT60 will turn harsh trumpet brightness into a reverberant shower of high frequencies that destroys mix clarity. The 3 kHz presence boost is where Chicago-style trumpet cut comes from — keep it precise with Black Knob.*

Ch 18 — Saxophone | Shure Beta 98H/C | EQ: Black Knob

Section	Parameter	Details
INPUT	Analog Sat	OFF — Keep sax clean for natural tone
FILTER	LPF	OFF — Sax has rich upper harmonics worth preserving
FILTER	HPF	120 Hz — Alto sax fundamental ~155 Hz; tenor ~98 Hz. HPF below this
EQ — HF	HF Shelf	8 kHz / -1.5 dB — Sax breathiness and reed noise can be harsh in reverberant rooms; small trim
EQ — HMF	HMF Bell	2 kHz / +2.5 dB / Q 1.2 — Sax presence and note definition; this is the cut-through frequency for sax in the horn section
EQ — LMF	LMF Bell	300 Hz / -3 dB / Q 0.8 — Sax body mud in reverberant room; same Memo treatment as all horns
EQ — LF	LF Bell	150 Hz / +2 dB / Q 0.5 — Sax warmth and body; especially important for tenor sax
COMP	Threshold	-20 dBFS
COMP	Ratio	3:1
COMP	Attack	SLOW — Sax attack is musical breath and reed articulation
COMP	Release	~250 ms auto
COMP	Makeup Gain	+2 dB
GATE	State	OFF
NOTE	Alto vs Tenor	If both alto and tenor are played, adjust HPF: Alto = 150 Hz, Tenor = 100 Hz; adjust LF boost accordingly

Engineer Notes: *Saxophone on the Beta 98H/C sits between trombone and trumpet in the frequency spectrum. The 2 kHz HMF boost is the sax's defining presence frequency — it's where sax cuts through a full horn section. The 300 Hz LMF cut is the universal Memo mud-control move for all horn channels. Body warmth at 150 Hz gives sax the chest resonance that makes it feel live and present. Note: if the performer switches between alto and tenor, adjust HPF and LF settings accordingly during the performance.*

Ch 19 — Flute | DPA 4099 (d:vote CORE) | EQ: Black Knob

Section	Parameter	Details
INPUT	Analog Sat	OFF — DPA 4099 is a reference-quality supercardioid condenser; no THD needed
NOTE	DPA 4099	Supercardioid pattern for superior gain-before-feedback; uniform off-axis response; 20Hz-20kHz; effective range 80Hz-15kHz with 2dB soft boost at 10-12kHz
FILTER	LPF	OFF — Flute needs all its harmonic content above 10 kHz
FILTER	HPF	200 Hz — Flute fundamental starts at ~261 Hz (middle C); aggressive HPF removes breath noise and stand rumble below the instrument
EQ — HF	HF Shelf	12 kHz / +1.5 dB — Enhance the DPA's natural 10-12 kHz soft boost; flute air and shimmer helps it cut through the mix
EQ — HMF	HMF Bell	4 kHz / +2 dB / Q 1.5 — Flute note articulation and clarity; this is where flute 'speaks' in a mix
EQ — LMF	LMF Bell	400 Hz / -3 dB / Q 1.0 — Flute body/hollow tone buildup in reverberant rooms; the DPA's close placement can accentuate this
EQ — LF	LF	FLAT — HPF at 200 Hz handles the low end
COMP	Threshold	-18 dBFS
COMP	Ratio	3:1
COMP	Attack	SLOW — Flute articulation is in the transient; let it speak
COMP	Release	~200 ms auto
COMP	Makeup Gain	+2 dB
GATE	State	OFF — DPA 4099 supercardioid rejects well; no gate needed

Engineer Notes: *The DPA 4099 is arguably the finest clip-on instrument mic in the world — its supercardioid pattern and reference-quality capsule make it ideal for flute, which is notoriously difficult to amplify with gain-before-feedback. Black Knob EQ with clean corrections respects the DPA's natural accuracy. The 200 Hz HPF is more aggressive than standard because flute fundamentals start at 261 Hz — everything below that is breath noise, key noise, and room rumble. The 12 kHz air boost enhances the DPA's natural 10-12 kHz soft boost characteristic, adding the flute's characteristic shimmer.*

Ch 20 — Keys 1 | DI (Active) | EQ: Black Knob

Section	Parameter	Details
INPUT	Analog Sat	ON — Light saturation (+2 dB drive); keys benefit from subtle analog warmth
FILTER	LPF	OFF — Keys are full range by nature
FILTER	HPF	50 Hz — Remove sub from keys that overwhelms the room
EQ — HF	HF Shelf	10 kHz / +1.5 dB — Keys air and brilliance
EQ — HMF	HMF Bell	3 kHz / +1.5 dB / Q 1.5 — Keys presence and attack
EQ — LMF	LMF Bell	250 Hz / -3 dB / Q 0.8 — Memo mud cut for keys; organ-style sounds especially bloom here
EQ — LF	LF Shelf	80 Hz / -2 dB — Gently reduce keyboard low-end that will bloom in Memo; organ pads can dominate the room
COMP	Threshold	-18 dBFS
COMP	Ratio	3:1
COMP	Attack	SLOW — Keys articulation is in the front of notes
COMP	Release	~300 ms auto
COMP	Makeup Gain	+2 dB
GATE	State	OFF

Engineer Notes: *Keys 1 is the primary keyboard channel — likely piano/organ/pad sounds that provide the harmonic foundation of the CTA tribute. Black Knob EQ is appropriate: precise corrections on keys rather than the Brown Knob's wider character. The 250 Hz cut is essential in Memo, where organ and synth pad sounds at this frequency will bloom into a wash in the 2.2s decay. Reduce the LF shelf slightly rather than HPF: keys need some low-end warmth but should not compete with bass guitar and kick in the sub range.*

Ch 21 — Keys 2 | DI (Active) | EQ: Black Knob

Section	Parameter	Details
INPUT	Analog Sat	ON — Light saturation
FILTER	LPF	OFF
FILTER	HPF	80 Hz — Keys 2 may have different range assignment; slightly tighter HPF
EQ — HF	HF Shelf	10 kHz / +1 dB
EQ — HMF	HMF Bell	2.5 kHz / +1.5 dB / Q 1.5 — Keys 2 presence, slightly different voicing from Keys 1
EQ — LMF	LMF Bell	300 Hz / -2.5 dB / Q 0.8 — Mud cut
EQ — LF	LF Shelf	100 Hz / -1.5 dB
COMP	Threshold	-18 dBFS
COMP	Ratio	3:1
COMP	Attack	SLOW
COMP	Release	~300 ms auto
COMP	Makeup Gain	+2 dB
GATE	State	OFF
NOTE	Assignment	Confirm with keyboard player what sounds are on Keys 2 vs Keys 1; adjust EQ if Keys 2 runs lead synth lines vs pads

Engineer Notes: *Keys 2 mirrors Keys 1 with slight EQ variations for differentiation in the mix. The most important action before soundcheck: confirm the keyboard player's sound assignments — if Keys 2 runs synth lead lines, the HMF presence boost should be more aggressive (up to +3 dB). If Keys 2 runs pads/strings, reduce the HPF to 50 Hz and soften the LMF cut. Both keyboard channels should have distinct spectral footprints to avoid frequency masking.*

Ch 22 — Keys Vocal | Shure Beta 58 | EQ: Black Knob

Section	Parameter	Details
INPUT	Analog Sat	ON — Light drive; Beta 58 benefits from subtle saturation to add warmth to its slightly bright character
FILTER	LPF	OFF — Vocal air is important
FILTER	HPF	120 Hz — Remove low-end proximity effect and stage rumble; keys player mic stands vary
EQ — HF	HF Shelf	10 kHz / -1 dB — Beta 58 has forward presence; small trim prevents harshness in Memo's reverb
EQ — HMF	HMF Bell	3 kHz / +2 dB / Q 1.5 — Vocal presence; Beta 58 needs a small nudge at 3k for cut-through in a dense mix
EQ — LMF	LMF Bell	250 Hz / -4 dB / Q 0.8 — CRITICAL Memo vocal mud cut; male vocals in this room build heavily at 200-300 Hz
EQ — LF	LF Bell	180 Hz / +1 dB / Q 0.6 — Small warmth restoration after the mud cut
COMP	Threshold	-20 dBFS
COMP	Ratio	4:1 — Background vocal needs tight dynamic control
COMP	Attack	SLOW — Let consonant attack through for intelligibility
COMP	Release	~200 ms auto
COMP	Makeup Gain	+3 dB
GATE	Threshold	-35 dBFS — Gate to reduce keyboard bleed when mic is not in use
GATE	Range	-40 dB
GATE	Attack	FAST
GATE	Release	200 ms

Engineer Notes: *The Beta 58 is a proven workhorse for live vocals — its tight supercardioid pattern and presence peak at ~6 kHz cut through band noise well. Black Knob EQ gives clean precise corrections. The 250 Hz LMF cut at -4 dB is the most critical move: male vocal mud at this frequency in Memo's 2.2s RT60 will smear vocal intelligibility, particularly problematic for background vocals that need to be heard without dominating. Gate to clean up keyboard bleed between vocal passages.*

Ch 23 — Bass Vocal | Shure Beta 58 | EQ: Black Knob

Section	Parameter	Details
INPUT	Analog Sat	ON — Light saturation for warmth
FILTER	LPF	OFF
FILTER	HPF	120 Hz
EQ — HF	HF Shelf	10 kHz / -1 dB
EQ — HMF	HMF Bell	3 kHz / +2.5 dB / Q 1.5 — Vocal presence; bass player may be further from the board than dedicated vocalists
EQ — LMF	LMF Bell	250 Hz / -4.5 dB / Q 0.8 — Memo mud; bass player vocal proximity + room = significant low-mid buildup
EQ — LF	LF Bell	180 Hz / +1 dB / Q 0.6
COMP	Threshold	-20 dBFS
COMP	Ratio	4:1
COMP	Attack	SLOW
COMP	Release	~200 ms auto
COMP	Makeup Gain	+3 dB
GATE	Threshold	-33 dBFS
GATE	Range	-40 dB
GATE	Attack	FAST
GATE	Release	200 ms

Engineer Notes: *Bass Vocal matches the Keys Vocal template with a slightly more aggressive 250 Hz cut (-4.5 dB vs -4 dB) because the bass player will have more proximity effect from the instrument's low frequencies near the vocal mic. The additional half-dB at 250 Hz prevents the bass instrument's low-end energy from being picked up by the vocal mic and creating mud. VCA group this with the other background vocals for consistent level management through the show.*

Ch 24 — Drum Vocal | Shure Beta 58 | EQ: Black Knob

Section	Parameter	Details
INPUT	Analog Sat	ON — Light saturation
FILTER	LPF	OFF
FILTER	HPF	140 Hz — Drum kit is the loudest thing on stage; more aggressive HPF on drum vocal to reject kit bleed
EQ — HF	HF Shelf	10 kHz / -1.5 dB — Drummer's vocal may have energetic delivery; extra high-end taming
EQ — HMF	HMF Bell	3.5 kHz / +3 dB / Q 1.5 — Presence needs to be strong to overcome drum kit noise pickup
EQ — LMF	LMF Bell	300 Hz / -5 dB / Q 0.8 — Most aggressive mud cut of all vocals; drum kit proximity creates significant 200-400 Hz buildup in mic
EQ — LF	LF Bell	200 Hz / +1 dB / Q 0.6
COMP	Threshold	-18 dBFS — Tighter control; drummer's vocal mic environment is chaotic
COMP	Ratio	6:1 — Most aggressive ratio of all vocals; drum kit dynamics are enormous
COMP	Attack	FAST — Control peaks from shouted phrases
COMP	Release	~150 ms auto
COMP	Makeup Gain	+4 dB
GATE	Threshold	-28 dBFS — Aggressive gate; must close hard between vocal phrases to reduce drum bleed
GATE	Range	-50 dB — Very hard close
GATE	Attack	FAST
GATE	Release	250 ms — Long enough to avoid chopping vocal tails

Engineer Notes: *Drum vocal is one of the most challenging channels in any live mix. The drummer is surrounded by the loudest sources on stage, and the Beta 58 will pick up significant drum bleed. The 300 Hz LMF cut at -5 dB is the most aggressive of all vocal channels because kick drum and snare proximity effect creates enormous low-mid buildup. The 6:1 compression ratio controls the drummer's tendency to vary vocal levels dramatically. The -50 dB gate range is intentionally aggressive — only open when the drummer is singing. Verify gate settings carefully during soundcheck.*

Ch 25 — Guitar Vocal | Shure Beta 58 | EQ: Black Knob

Section	Parameter	Details
INPUT	Analog Sat	ON — Light saturation
FILTER	LPF	OFF
FILTER	HPF	120 Hz
EQ — HF	HF Shelf	10 kHz / -1 dB
EQ — HMF	HMF Bell	3 kHz / +2.5 dB / Q 1.5 — Guitar player vocal presence
EQ — LMF	LMF Bell	250 Hz / -4 dB / Q 0.8 — Standard Memo vocal mud cut
EQ — LF	LF Bell	180 Hz / +1 dB / Q 0.6
COMP	Threshold	-20 dBFS
COMP	Ratio	4:1
COMP	Attack	SLOW
COMP	Release	~200 ms auto
COMP	Makeup Gain	+3 dB
GATE	Threshold	-33 dBFS
GATE	Range	-40 dB
GATE	Attack	FAST
GATE	Release	200 ms

Engineer Notes: *Guitar Vocal is a standard background/supporting vocal channel using the same Beta 58 template as Keys Vocal and Bass Vocal. Consistent processing across all background vocal channels creates a cohesive sound for group harmony passages, which are central to the Chicago tribute style. Ensure all background vocal channels are in the same VCA group so they can be pushed up together during vocal-heavy sections.*

Ch 26 — Star Vocal (Lead) | Shure Beta 58 | EQ: Black Knob

Section	Parameter	Details
INPUT	Analog Sat	ON — Moderate drive (+3 dB); the lead vocal through Beta 58 with SSL saturation is a classic combination
FILTER	LPF	OFF
FILTER	HPF	100 Hz — Lead vocal gets slightly looser HPF than backgrounds; more body wanted
EQ — HF	HF Shelf	12 kHz / +1 dB — Lead vocal air and presence; singer should sound open and forward
EQ — HMF	HMF Bell	3.5 kHz / +3 dB / Q 1.2 — Lead vocal presence; needs to cut through full band + horn section in Memo's acoustic environment
EQ — LMF	LMF Bell	230 Hz / -5 dB / Q 0.8 — MOST CRITICAL CUT in the show; lead vocal at this frequency in Memo's 2.2s RT60 will completely wash out unless aggressively controlled
EQ — LF	LF Bell	180 Hz / +1.5 dB / Q 0.6 — Vocal body restored after the mud cut
COMP	Threshold	-22 dBFS — Most sensitive threshold; lead vocal is always in the compressor
COMP	Ratio	3:1 — Moderate ratio; lead vocal dynamics are musical and should not be over-squashed
COMP	Attack	SLOW — Preserve vocal consonant attack for intelligibility
COMP	Release	~150 ms auto — Faster release than backgrounds; lead vocal should feel energetic
COMP	Makeup Gain	+3 dB
COMP	Routing	Dyn pre-EQ — Compress the raw vocal for consistent delivery before EQ
GATE	Threshold	-38 dBFS — Looser gate threshold; lead vocal should open easily
GATE	Range	-20 dB — Partial close; don't hard-gate the lead vocal
GATE	Attack	FAST
GATE	Release	250 ms

Engineer Notes: *The lead vocal is the most important channel in the show — it must be intelligible, present, and emotionally compelling over a 20+ piece ensemble in a reverberant 556-seat space. The 230 Hz LMF cut at -5 dB is the most critical single EQ move in the entire show: this is where Memo's RT60 hits the lead vocal hardest, washing out the fundamental and creating a boomy, indistinct sound. The 3.5 kHz HMF boost at +3 dB cuts through the dense texture. Use a -20 dB gate range (partial close) so the vocal mic breathes naturally. During the show, this fader should have your right hand on it at all times.*

GLOBAL FX — Seventh Heaven Pro (Bricasti M7 Emulation) — Memorial Hall

MEMO-SPECIFIC REVERB PHILOSOPHY: Memorial Hall already has a natural RT60 of ~2.2 seconds. Seventh Heaven Pro should be used CONSERVATIVELY — mostly for dimensional placement, not reverb augmentation. The room itself is your reverb. Send levels should be 8-12 dB lower than you would use in a dry room.

Bus / Instrument	Parameter	Setting / Details
VOCAL VERB (Lead/BG vocals)	Preset	7H Pro — 'Studio B' or 'Control Room C' — small room verb for intimacy, NOT a large hall preset. The room provides the hall.
VOCAL VERB	Pre-Delay	25-30 ms — Creates separation between dry vocal and reverb tail; prevents smearing in the room
VOCAL VERB	Decay Time	0.8 – 1.1 s — Intentionally SHORT; the natural room adds another ~1.5s on top
VOCAL VERB	HF Damping	2.5 kHz @ -6 dB — Dark, warm reverb tail; bright reverb tails become harsh in Memo
VOCAL VERB	Send Level	-18 to -16 dBFS from vocal channels — Very conservative; just enough for depth
HORN VERB (Tromb/Trumpt/Sax)	Preset	7H Pro — 'Studio C' or 'Wood Room' — warm, tight diffusion that enhances brass without wash
HORN VERB	Pre-Delay	15-20 ms
HORN VERB	Decay Time	0.6 – 0.8 s — VERY short; horns in Memo's reverb field are already enveloped by the room
HORN VERB	HF Damping	3 kHz @ -4 dB
HORN VERB	Send Level	-20 to -18 dBFS — Use only for cohesion between horn mics, not for reverb effect
KEYS/GUITAR VERB	Preset	7H Pro — 'Small Stage' or 'Studio A' — similar to vocal verb but slightly larger
KEYS/GUITAR VERB	Pre-Delay	20 ms
KEYS/GUITAR VERB	Decay Time	1.0 – 1.3 s
KEYS/GUITAR VERB	Send Level	-22 dBFS — Keys and guitar in Memo need very little additional reverb
GLOBAL	IMPORTANT	DO NOT send drum channels to reverb in Memo. The room's 2.2s RT60 provides all the drum ambience you need. Adding reverb to drums will make them sound like they're in a cavern.
GLOBAL	Monitoring	Always check wet/dry balance in mono with Smaart v9 active to verify you're not building room mode resonances with reverb tails

CRITICAL NOTES — Memorial Hall • Cincinnati Transit Authority 2026

MEMO RT60 REMINDERS

Memorial Hall's 2.2s RT60 is the defining factor in ALL processing decisions on this sheet. Every mud cut, every conservative reverb send, every aggressive HPF — all calibrated for this room. If you are used to mixing in drier rooms, your instinct will be to add reverb, boost low-mids, and be gentle with mud cuts. Fight that instinct. The room is already doing those things for you.

SSL EV2 BROWN vs BLACK KNOB

Brown Knob (O2): Use for drums, bass, horns, electric guitar — instruments that benefit from the wider, grittier SSL reshaping. Black Knob (242): Use for all vocals, acoustic instruments, flute, keys — cleaner, more precise corrections. This is not arbitrary: the EV2 models two distinct historical SSL EQ circuits and they sound genuinely different.

EV2 ANALOG SATURATION

Analog ON adds THD (Total Harmonic Distortion) at both input and output stages, emulating the SSL 4000E console's preamp character. For clean sources (flute/DPA, overheads/SR20, playback tracks): ANALOG OFF. For everything else: ANALOG ON with appropriate input drive. Driving the input 2-5 dB beyond unity creates progressively more harmonic richness — great for bass, drums, guitars.

SUGGESTED VCA GROUPS

Group 1 — DRUMS: Ch 1-9 (Kick, Snare, Hat, Toms x3, OH L/R, Knee) | Group 2 — BASS: Ch 11-12 (DI + AMP linked) | Group 3 — GUITARS: Ch 13-15 | Group 4 — HORNS: Ch 16-19 (Trombone, Trumpet, Sax, Flute) | Group 5 — KEYS: Ch 20-21 | Group 6 — BG VOCALS: Ch 22-25 (Keys/Bass/Drum/Guitar vocals) | Group 7 — LEAD VOCAL: Ch 26 (Star Vocal — solo fader control) | Group 8 — PLAYBACK: Ch 10

WORKFLOW — SOUNDCHECK ORDER

Recommended order: 1) Kick & Snare (establish low-end floor) | 2) Overheads (check phase with Smaart) | 3) Toms | 4) Bass DI+AMP blend | 5) Guitars | 6) Full Band — check horn section in context | 7) Lead Vocal last — dial in around the full band, not in isolation. DO NOT dial in lead vocal in a quiet room and expect it to translate when the full band is playing in Memo.

SMAART v9 CHECKPOINTS

Run Smaart v9 at these key moments: 1) Pink noise sweep pre-show to identify Memo's room modes (expect nodes around 63 Hz, 125 Hz, and 250-315 Hz in this room size). 2) OH Left/Right phase alignment check with close mics before drum soundcheck. 3) Full band RTA during soundcheck to verify 200-400 Hz mud control is working. 4) Lead vocal intelligibility check — STI (Speech Transmission Index) target >0.60 in this room.

HORN SECTION — SPECIAL NOTE

Chicago's horn section is the signature element of the CTA tribute. Trombone (Ch 16), Trumpet (Ch 17), Sax (Ch 18) should be treated as a unified section. During soundcheck, bring the full horn section up together and check their blend. 300-350 Hz is the shared mud zone — if any horn sounds boomy, increase the LMF cut. The horns should sound bright, present, and clean — NOT reverberant or washy in Memo.

BASS DI + AMP BLEND

Ch 11 (Bass DI) and Ch 12 (Bass AMP) should be blended on a single VCA/DCA. Starting blend: DI 70%, AMP 30%. The DI provides punch and definition; the AMP provides warmth. In Memo's reverberant environment, keep bass mid-forward (800 Hz-1.5 kHz) to maintain definition through the room's decay. A bass that's too sub-heavy will bloom in the room and become undefined.

FLUTE — DPA 4099 GAIN MANAGEMENT

The DPA 4099's supercardioid pattern is excellent for gain-before-feedback — but flute is still the hardest instrument in this show to amplify without feedback. Set the flute channel last during soundcheck. Ring it out carefully in Memo before the show. If feedback occurs, first check the 200-400 Hz range in the room; then check 4-5 kHz. The DPA's 10-12 kHz soft boost can excite monitor feedback if wedge levels are too high.
